Automated Microdrama Promo Discovery & Ranking System

Solution Document (Problem 1) • Show: *I Married a Broken Soldier. He was a Duke* (50 Episodes) • Harman Singh

1. Problem Definition & Editorial Assumptions

Business Objective: Microdrama platforms acquire >80% of top-of-funnel viewers through 20–60s vertical clips on TikTok, YouTube Shorts, and Instagram Reels. Manually analyzing 50+ screenplay episodes per series is an expensive editorial bottleneck. This pipeline automates the extraction and ranking of high-converting promo moments from timecoded scripts with cut-boundary safety and cold-audience viability.

- **Timestamp Fidelity:** Production script timecodes correspond accurately to master video shot transitions.
- **Safe Cut Invariant:** CAMERA shot-start events represent clean visual cut boundaries without tearing audio.
- **Dialogue Primacy:** Spoken dialogue provides primary emotional signals; visual-only actions receive safety weighting.
- **Top-of-Funnel Conversion:** Promos must terminate on unresolved open loops; end-of-series spoilers are penalized.

2. End-to-End Pipeline: AI Perception + Deterministic Enforcement

Production reliability is achieved by strictly separating **creative narrative search (LLM)** from **mechanical rule enforcement (Code)**:

Stage	Component / Owner	Role, Logic & System Invariants
1. Parse	Deterministic Code promo/parser.py	Ingests screenplay blocks (CAMERA, ACTION, SOUND FX, DIALOGUE). Assigns immutable, sequential IDs (e.g. E3-41) and indexes camera shot boundaries.
2. Propose	LLM (gemini-3.5-flash-lite) promo/llm.py	Examines 2-episode sliding windows. Identifies dramatic conflicts and cross-scene stitched clips. **Core Constraint: Model outputs ONLY event IDs; timestamps are forbidden**.
3. Validate	Deterministic Code promo/validate.py	Resolves IDs to precise timestamps; snaps cuts to shot boundaries; expands cuts that split spoken lines mid-sentence; strictly discards clips <20s or >60s.
4. Cold View	Secondary LLM promo/llm.py	Inspects isolated clip text with zero context or loglines (blind TikTok scroller persona). Rates standalone comprehension (1–5) and confirms unresolved tension.
5. Rank	Deterministic Code promo/rank.py	Calculates weighted composite score; eliminates near-duplicates (Temporal IoU > 0.40); enforces diversity cap of max 2 promos per episode in top cohort.

3. Multi-Factor Scoring & Ranking Formulation

Composite Score = 0.35 × Hook + 0.35 × Cliffhanger + 0.20 × Standalone + 0.10 × EarlyEpisode – 0.15 × SpoilerPenalty + TailBonus

- **Hook (35%):** First 3 seconds dictate swipe rate. Scores physical actions (slaps, grabs) and high-voltage betrayal lines.
- **Cliffhanger (35%):** Terminal second cuts on unfulfilled questions or threats, never after an apology or resolution.
- **Standalone Clarity (20%):** Cold-viewer score validating that a stranger understands character stakes without lore.
- **Early Episode & Tail Bonus (+0.4):** Prioritizes early episodes (landing page conversion) and script-authored ending hooks.

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4. Quantitative Telemetry: 50-Episode Benchmark & Validation Dynamics

The full pipeline was executed across all 50 screenplay episodes of *I Married a Broken Soldier. He was a Duke*:

Total Episodes Analyzed 50 Episodes (Full Series)	**Total LLM API Calls** 70 Calls (49 prop / 21 cold)	**Candidate Proposals** 161 Raw Proposals	**Valid Clips Accepted** 71 Clips (44.1%)
Rejected by Validation 90 Clips (55.9% strict safety)	**Pipeline Execution Time** 500.7s (~8.3 min total run)	**Average Top 10 Duration** 28.5 seconds (20.5s–42.5s)	**API Model & Endpoint** gemini-3.5-flash-lite

Top Discovered Promo Clips (Curated Cohort):

Rank / Target	Dur.	First 3s Hook Dynamic	Terminal Cliffhanger Cut
#1 Ep1 (01:57–02:18)	21.0s	Heroine spots groom holding sister's hand at wedding altar.	Groom announces cancellation to cathedral; freezes on bride's tear.
#2 Ep38 (00:55–01:27)	32.4s	Juliana turns to dungeon wall, vowing loyalty to husband.	Execution wedding bells chime; abrupt blackout on impending gallows.
#3 Ep1 (00:32–01:03)	30.2s	Matriarch cold line: *"Gerald is not waiting for you anymore."*	Juliana directly confronts groom: *"Why are you holding her hand?"*
#5 Ep3 (01:00–01:25)	25.5s	Guards aggressively seize abandoned bride by both shoulders.	Scarred stranger drops guard with one punch and kneels with ring.

5. Production Economics & Editorial Workflow

- **Token & Cost Efficiency:** A 50-episode season consumes ~350k input tokens. At Flash-tier pricing, total inference cost is **under $0.05 per entire 50-episode show**, fitting entirely within free tier developer quotas.
- **Horizontal Throughput:** Independent show pipelines allow parallel cloud execution; a batch of 100 titles processes in under 30 minutes across asynchronous workers.
- **Human-in-the-Loop Efficiency:** Eliminates manual scanning. Editors review the top 10 ranked cards in the Streamlit UI, reducing human turnaround from 3 hours to under 5 minutes per title.

6. Tech Stack & Critical AI Failure Analysis (Assignment Requirement)

Implementation Stack: Python 3.11, OpenAI SDK, Streamlit Community Cloud, PyYAML, ReportLab.

Real-World Failure Observed in Run: LLMs suffer from severe *duration hallucinations* when interpreting screenplay timecodes. For example, in candidate **P006** (Ep3 church rescue), the AI correctly identified the narrative climax but proposed an entire **151.4-second continuous sequence** (>2.5× the 60s hard ceiling), which would cause instant rejection on TikTok. In candidate **P013**, it selected a single line of dialogue lasting only **19.5s** (<20s ceiling).

Defensive Engineering Remediation: Because prompt instructions alone cannot reliably constrain LLM arithmetic on timecodes, duration enforcement was offloaded entirely to deterministic code in promo/validate.py. Furthermore, enabling multi-segment event stitching allowed the pipeline to isolate just the punch and proposal moments into a compliant **42.3s clip** without editorial intervention.

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